

AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

**MODEL AND CONTROLLER DESIGN
FOR A NOVEL SYNCHROPTER
CONCEPT FOR THE AERIAL-RECOVERY
OF FALLING ROCKET STAGES**



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Submission Date: 15/06/2020

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Abstract

A model for payload dynamics is conducted and presented, and interaction between helicopter and payload modes are evaluated. Additionally, a LQG controller is proposed to improve the open loop stability performance of the synchropter

WRR Team Members

Table 1: *Team Members and Roles.*

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Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
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†Sub-team leads are shown in **bold**

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1 Preliminary work: Selecting recovery vehicle

The first phase of the design project consisted of deciding on an appropriate vehicle for recovery. Feasibility of vertical take off and lift (VTOL) vehicles were considered from control architecture design point and safety point of view. A tiltrotor/air crane design configuration was proposed in collaboration with other team members. With tiltable engines present near wing tips, the proposed design had the benefits of cruise efficiency of fixed wing aircrafts with hover capabilities of a helicopter.

For flight mechanics analysis, the VTOL would be modelled as a fixed wing aircraft during cruise and as a helicopter during takeoff, landing, and hover flight phases. To ensure low pilot workload during remotely controlled sessions, an effective control mixing system must be present to enable use of fixed wing aircraft and helicopter control surfaces which increases complexity of FCS design. Finally, while dynamics during cruise or hover can be captured accurately with linear model, the tilting phase of the rotors must be modelled with a time variant system and requires applications of non-linear control methods. Hence, system modelling and control design for a VTOL is far more complicated than for either a helicopter or fixed wing aircraft.

2 Introduction

This technical report aims to highlight a part of the contribution made by the Flight Mechanics, Control and performance team to the design of a synchropter purposed for aerial recovery. The WRR05 team was responsible for modelling helicopter and payload dynamics, identifying and determining stability characteristics of the model, designing a controller and finally evaluating the helicopter's performance.

This report is split into three sections: Determination of inertial forces on rotor blades, modelling payload dynamics and methodology used to design a LQG flight controller for different mission legs. The reader is advised that due to the high level of collaboration between members of the team, cross-referencing will be present. Where required, reports written by other members with essential information has been cited.

3 Rotor Dynamics: Inertial Forces

An accurate model is essential for predicting stability characteristics accurately. Hence, rotor dynamics were included in the first iteration of the linearized state space model. This section highlights how geometric and material properties of the blade are used to determine its inertial forces and

moments, which is then combined with aerodynamic forces to compute the net rotor force and moment at helicopter centre of gravity.

3.1 Model assumptions

Inertial forces measure forces imparted on the rotor due to its acceleration and gravitational forces. Resultant inertial forces and moments F_I and M_I on the blades are [2]:

$$F_I = \int_A \rho(a_p + g\mathbf{k})dA$$

$$M_I = - \int_A \rho(y_0\hat{e}_x + z_0\hat{e}_y) \times (a_p + g\mathbf{k})dA$$

Where a_p is the acceleration vector of a local blade cross section element. The coordinates $\hat{e}_x$ and $\hat{e}_y$ represents the position of a particle along the local blade cross section as demonstrated in [2]. Though realistically aeroelastic deflections will be present, they add a lot to complexities in linear time invariant models and hence has been neglected. It is further assumed that all blades have the same geometric and material properties and are made from an isotropic material. No blade sweep has been considered, and finally the blades are assumed to be rotating at a constant rpm Ω at any given mission leg. The methodology used for determining forces, including derivations of term a_p has been highlighted in detail in [2].

3.2 Resultant Rotor Forces

Once a_p is determined, F_I and M_I are determined for along discretized blade cross sections. These forces are then added to the aerodynamic forces, and then integrated along the blade length to determine net rotor forces and moments:

$$F_R = \int_e^1 (F_A + F_I)dx$$

$$M_R = \int_e^1 (M_A + M_I)dx$$

Where the subscript A refers to aerodynamic components and e is the hinge offset distance from rotor hub. Coordinate transformations are applied [2], and the resultant Rotor forces F_{MR} and M_{MR} are calculated at helicopter centre of gravity.

4 Payload Dynamics

4.1 Motivation

As highlighted in the mission profile in [3] and based on the recovery system design [4], after retrieving the rocket stages, the helicopter will cruise back to a landing site with an externally suspended payload. The inclusion of slung load dynamics in determining helicopter stability isn't a novel concept, helicopters are often required to transport dangling payload for military operations. Incorporating payload dynamics in our existing linearised model is crucial as movement of the payload can have destabilising effects during hover and cruise phases. Typically, if the payload to helicopter gross weight ratio exceeds 30%, a high level of coupling between payload and helicopter dynamics is present [1] often resulting in residual oscillations. Around 10% of helicopter accidents

can be related to externally suspended payload operations [5], and payload dynamics also puts a maximum limit on cable length and vehicle cruise speed before oscillations diverge. Considering the high payload to helicopter mass ratio and a minimum cruise speed requirement for the possibility of using one vehicle for recovering multiple stages, payload dynamics cannot be neglected.

4.2 Slung load model

Several models with varying levels of complexity have been proposed in the past for slung load applications. For the purpose of the following analysis, the external payload is modelled as point mass suspended with a cable at a single point on the helicopter. Motion of the payload is very similar to that of a spherical pendulum, indicating that the payload can oscillate in two planes and has two degrees of freedom θ_L and ϕ_L . This is presented in 1.

Aerodynamic forces on the payload influence payload dynamics and create an additional moment on the helicopter. For the purposes of this model, the payload is assumed to be a non-lifting surface. However, payload drag is taken into consideration. On the other hand, grapple and cable drag is assumed negligible. Cable mass is further neglected as it is roughly 1% of the payload mass. Finally, the cable is assumed to be inelastic. This was decided upon considering the high cable stiffness and the relatively small effects of cable elasticity in single suspended payload operations [ref NASA]. The limitations of this model and aforementioned assumptions have been presented in section 4.5

4.3 Slung load Dynamics

The body frame of reference is at the helicopter CG, denoted with subscript **B**. The suspension frame of reference (**H**) is parallel to body frame of reference but originates at the suspension point. In this configuration, when perturbed, suspended load will apply a force and moment (F_{PL} and M_{PL}) at the suspension point along an inextensible cable of length L . Payload position R_L , with respect to suspension is given as:

$$R_L = -L\sin(\theta_L)\cos(\phi_L)i_H + L\sin(\theta_L)\sin(\phi_L)j_H + L\cos(\theta_L)k_H \quad (1)$$

Additionally, distance of suspension point from helicopter CG is:

$$R_H = x_A i_H + y_A j_H + z_A k_H \quad (2)$$

Position R of payload with respect to helicopter CG is hence evaluated as $R = R_H + R_L$.

The absolute payload velocity V_L will depend on helicopter's translational and angular velocity vectors $V_{CG} = ui_B + vj_B + wk_B$ and $\omega = pi_B + qj_B + rk_B$ respectively:

$$V_L = V_{CG} + \dot{R} + \Omega \times R \quad (3)$$

Finally, payload acceleration a_L is obtained as:

$$a_L = \ddot{V}_{CG} + \ddot{R} + \dot{\Omega} \times R + 2\Omega \times \dot{R} + \Omega \times (\Omega \times R) \quad (4)$$

The cable tension experienced by the helicopter at suspension point depends on a_L , gravitational forces F_G on payload and finally, payload drag D_{PL} . Even though the payload is considered to be a spherical mass for stability analysis purposes, payload drag during different segments of the flight has been calculated by modelling the rocket stage as a cylinder in [6]. F_{PL} and M_{PL} at suspension point is given as:

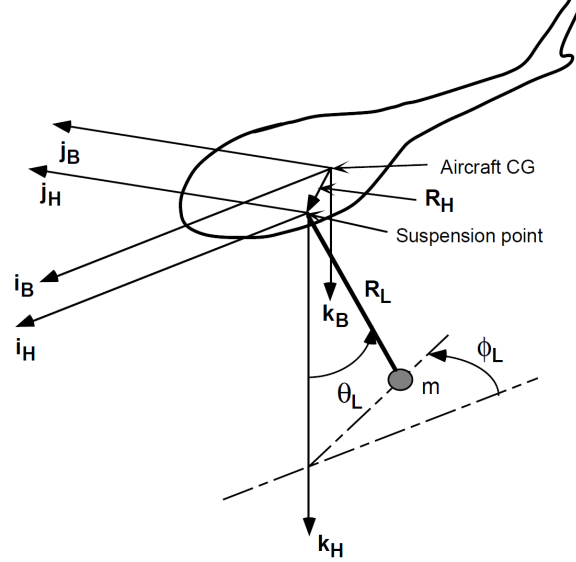


Figure 1: Illustrates point mass m , payload degrees of freedom and body and hook frames of reference [1]

$$F_{PL} = ma_L + F_G + D_{PL}$$

$$M_{PL} = F_{PL} \times R_H$$

F_{PL} and M_{PL} are resolved into components X, Y, Z and L, M, N respectively, added to the fuselage forces [7] and then integrated in the linearised state space model [8] to obtain a combined helicopter and slung load model. To complete this model however, equations of motion of payload d.o.f (θ_L and ϕ_L) are needed. These are obtained by enforcing a moment equilibrium at the suspension point, resulting in:

$$R_L \times (ma_L + F_G + D_{PL}) = 0 \quad (5)$$

From equation 5, second order differential equations for θ_L and ϕ_L are obtained. These equations consist of several non-linear expressions, and coupling between helicopter and payload states. The expressions are linearised at equilibrium points for different mission legs and added to the state space model in [8].

4.4 Effects of Payload motion on Dynamic Stability

Dynamic stability of the helicopter-payload system for $m_L = 554\text{Kg}$, $L = 36\text{m}$ during hover and $u = 30\text{m/s}$ during cruise has been conducted in detail in [8]. This section focus on the payload modes and its interaction with helicopter modes during these missions legs.

The eigenvalues of payload modes are marginally stable [8], indicating that there is no natural system damping. Contribution of lateral and longitudinal oscillations due to perturbations away from trim points for each mode is identified with eigenvector analysis. During hover, oscillations in the lateral plane are far more dominant than oscillations in the longitudinal planes. Longitudinal plane payload modes aren't clearly identifiable. Additionally, some coupling can be observed between the lateral payload modes and the helicopter's dutch roll and spiral modes. While payload oscillations clearly affect helicopter's natural modes, no significant effects of helicopter's motion are observed on

payload modes. During cruise, lateral and longitudinal oscillation modes can be clearly identified and there isn't significant coupling between them. Contribution of longitudinal payload states is observed in helicopter's pitch subsidence mode, and coupling between dutch roll and lateral oscillation mode decreases.

Due to the results obtained, a study of effects of slungload properties on system stability is further conducted. This not only aids recovery system and control design, but also a flexibility in choice of load mass, cable length and maximum cruise velocity with payload enables use of synchropter for other missions. In this analysis, emphasis is placed on the impact of varying cable length and cruise velocity on the open loop stability of the combined synchropter/slung load system.

4.4.1 Cable Length

System stability during hover is evaluated at cable lengths 15m, 25m and 36m. Initially the first stage rocket was used for analysis, but after the simulation team demonstrated difficulties in hovering with this mass, the lighter payload mass was used instead. Figure 2b focuses on system poles near the origin, while figure 2a presents all system poles.

As visible in figure 2a, heavily damped, oscillatory stable and oscillatory unstable flight modes occur but in general, altering cable length doesn't have a substantial effect on helicopter modes. Varying cable length does have an effect on payload modes, which are represented by eigenvalues on the imaginary axis in figure 2b. The natural frequency of oscillation of the payload reduces with increasing cable length. This is expected as w_n of a simple pendulum is inversely proportional to string length. Eigenvector analysis further shows that increasing cable length helps stabilise payload oscillation in the longitudinal plane. Interestingly, highly coupled longitudinal and lateral payload modes can be observed at smaller cable length, but as length increases, longitudinal oscillations dissipate. Furthermore, as L increases to 36m, a bifurcation can be observed, resulting in oscillatory stable and unstable solutions. As this is the cable length selected for the first design iteration, use of a winching system to reduce cable length upon capture is recommended to improve open loop stability performance of the helicopter. Obtained results also indicate stabilisation of payload motion in the plane is required during hover, which can be achieved with optimal control algorithms such as LQR.

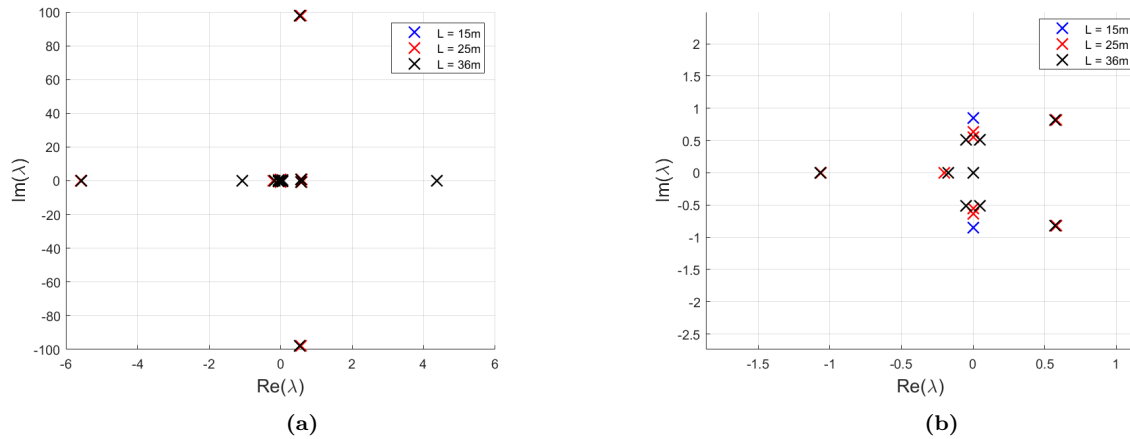


Figure 2: Shear force distribution over wing half span for symmetric loading case.

4.4.2 Cruise Velocity

System stability during cruise with first stage payload mass of 1400Kg is conducted at velocities 20m/s, 30m/s and 40m/s. The equilibrium payload angle θ_L is evaluated in [4] with the help of drag analysis conducted by [6]. Unlike cable length, altering cruise velocity does have visible effects on helicopter modes.

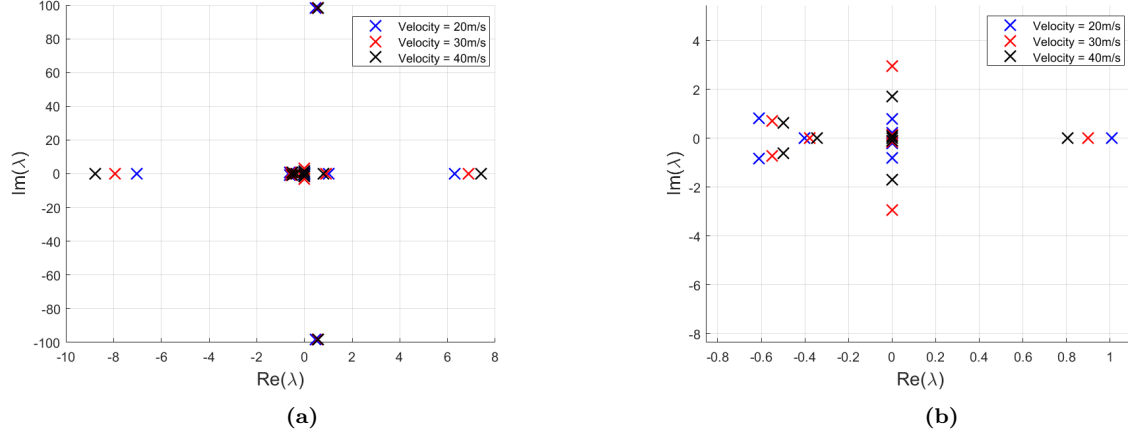


Figure 3: Shear force distribution over wing half span for symmetric loading case.

In general, damping of helicopter's pitch subsidence mode increases while eigenvalues associated with spiral mode move further down the right half plane. Phugoid and heave subsidence modes move closer towards the imaginary axis but no bifurcations were observed. The point of interest here however is to evaluate the contribution of payload dynamics to these changes. Eigenvector analysis of helicopter's mode show a decoupling between payload dynamics and lateral helicopter dynamics such as dutch roll motion. However, dominance of longitudinal oscillations can be observed in pitch subsidence mode with an increase in velocity. Interestingly, coupling between payload longitudinal motion and phugoid mode reduces with increasing velocity. Overall, no strong destabilising trends can be observed on the helicopter due to payload dynamics as velocity increases.

Coupling between longitudinal and lateral payload modes decreases with increasing velocity. Oscillation frequency reduces in the lateral plane, and there is an interesting trend observed in figure 3b, where oscillation frequency increases at 30m/s cruise speed, but reduces again at 40m/s. However, no unstable modes are identified. These results motivate stabilisation of payload motion in the longitudinal direction at higher cruise speeds. The general trends observed in system stability with increasing velocities are consistent with results obtained by the simulation team (cite Hakan), but this study can be extended to test system behaviour at higher velocities to validate its stabilising effects.

4.5 Model evaluation

The spherical pendulum model allows a reasonable analysis of payload dynamics for the first design iteration. However, there are several limitations to this model. Firstly, the complex parafoil, drogue chute, and rocket stage system has been simplified to a long cable with a point mass. This fails to account for any aerodynamic instabilities due to the payload shape [1]. Moreover, moments of inertia of the payload aren't modelled, hence effects of rotation of the payload about its stability axes due to any perturbation aren't considered. For first stage recovery, due to the high payload

mass and long cable length, the moment imparted on the helicopter due to these motions isn't non negligible. Moreover, assuming cable inelasticity neglects the vertical bounce motion of the payload. This phenomenon is important for high payload to vehicle mass ratio systems, and occurs when load oscillation natural frequency matches rotor natural frequency. Cable elasticity was neglected in this analysis due to difficulties in capturing cable slack in linear time invariant models. Finally, the obtained equations of motion consist of several non-linear and coupled terms. These go to zero as the system is linearized and evaluated about a trim point, however prediction of system dynamics at perturbations away from trim points may be inaccurate. This issue can be resolved with time variant models.

For the next design iterations, two improved models are suggested. First, the payload and helicopter can be modelled as a multibody system while still assuming cable inelasticity. The payload is hence modelled as a 6 degree of freedom system. The equations of motion for this system can be derived using Lagrangian mechanics, or Udwadia Kalaba equations [5]. The second proposition is to relax the inelastic cable assumption, and model the cable extension with a spring stiffness and damping ratio, thus accounting for cable tension only when there is an extension. While this results in the most accurate representation of the model, it requires nonlinear control methods, poses challenges in state measurement for controller design and greater computational expense.

5 Controller Design

The initial trade-off study conducted and mission aims considered to select Linear Quadratic Gaussian as the preferred Control method is presented in [9]. Considering the mission profile [3], limitations of linear time invariant control systems and destabilising effects of the payload, the flight controller will be tuned for two mission legs: Cruise with payload and hover with payload.

5.1 Linear Quadratic Regulator

The helicopter-payload system is modelled as a linear time invariant system of the form:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t)\end{aligned}$$

LQR is an optimal control method relying on full state feedback and is compatible with continuous LTI systems where feedback gains K are calculated by solving the Riccati equation. LQR can be applied as long as system (A,B) is stabilisable. To compute gains K , system cost is calculated with weighting matrices Q and R , where Q penalizes perturbation of states from trim point and R penalizes control input u . Relative sizing of Q and R provides the choice of either cheap or expensive control, which is quite useful in preventing actuator saturation. For this analysis, cheap control is preferred as stabilisation of system and reducing settling time is more important than control effort. However, the minimum weights selected for R are still limited by actuator saturation bounds.

Realistically, full state feedback is not possible, but state estimation can be performed. A sensible choice is using Kalman filters. Along with state estimation, Kalman filters also enable modelling of process and sensor noise by including zero mean, gaussian noise in the analysis, thus in theory improving the controller's robustness performance. Design of Kalman filter covariance matrices and noise modelling is conducted in [?]. Combination of LQR and Kalman filter state estimation is known as linear quadratic gaussian control.

5.1.1 Controllability

Before LQR can be applied, system controllability must be evaluated. This is done using Popov-Belevich-Hautus test [10] which states that system(A,B) with n states and n eigenvalues λ is controllable only if:

$$\text{rank}(\lambda I - A, B) = n$$

PBH tests conducted at designed equilibrium points for hover and cruise demonstrates that the system is controllable, and thus stabilisable, during both mission legs. This validates the use of full state feedback for controller design if observability can be assumed.

5.1.2 Tuning for Hover

For the first iteration, Q and R weights are selected by a combination of Bryson's rule [11] and review of the open loop stability of the system. For a system with n states and $i = 1, 2 \dots n$ Bryson's rule recommends creating a diagonal matrix Q where $Q_{ii} = \frac{1}{\max(x_i^2)}$, and x_i is the selected trim value for each state. Theoretically, the trim value for all states would be zero during hover, a small operating range has been specified and presented in table 5:

Bryson's rule ensures that states with smaller trim values are penalized more. Based on the selected operating range, this would penalize lateral motion and ensure attitude control. However, as open loop stability analysis is available, these weights can be refined to ensure that states contributing to unstable modes are penalized more. Based on the analysis conducted in [8], phugoid and spiral modes are unstable and in section 4.4, it was deduced that lateral plane oscillations have to be stabilised. Hence, Q weights for states u, v, p, r and ϕ_L were further penalized. As a starting point, all R weights are set to 0.1. Relative size of R weights is kept the same as no drastic differences are known in the saturation limit of actuators. The initial Q weights have been presented in 5, and the corresponding closed loop stability and payload states response is presented figures 4a and 4b. Overall, all systems are shown to be stable and a substantial improvement is observed in frequency of oscillation of the dutch roll modes when compared to open loop analysis [8]. However, some modes are highly damped suggesting that Q can be lowered. Additionally, while lateral payload states converge to equilibrium quickly, a large overshoot is observed which exceeds the maximum operating range for ϕ_L in hover. Based on these observations, and in combination with estimations made by the Kalman filter, Q can be further tuned till satisfactory response is observed.

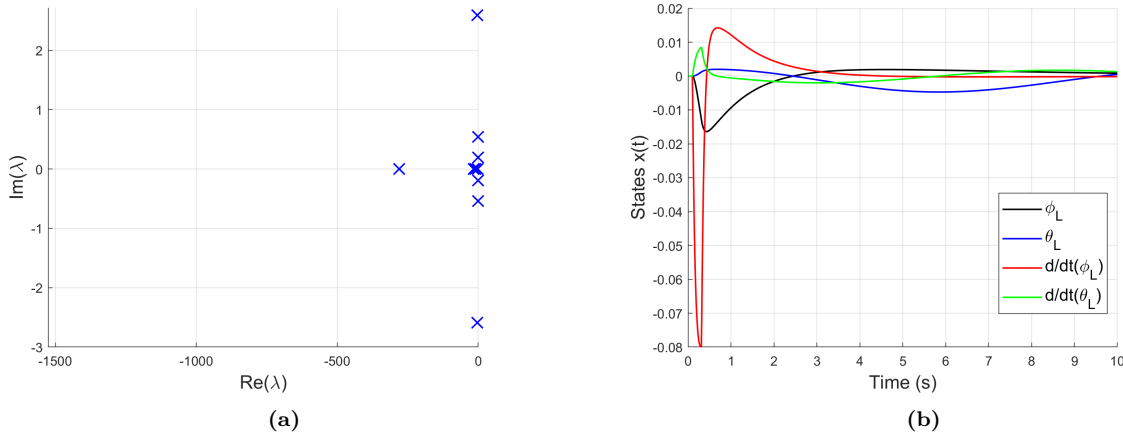


Figure 4: Shear force distribution over wing half span for symmetric loading case.

5.1.3 Tuning for Cruise

Same R is used as in hover and Bryson's rule is implemented again to obtain initial Q weights. Following the procedure above, open loop unstable modes and states contributing to these modes are identified. For cruise, unstable helicopter modes consist of phugoid, spiral and roll subsidence modes, and longitudinal oscillation mode for the payload. Hence, weights Q weights corresponding to states u, v, p, r and θ_L are changed and presented in 5. System and stability response is plotted and presented in figures 5a and 5b. Again, though system is stable, high damping is present and system response shows high overshoot and oscillations in longitudinal payload states before convergence. Moreover, a steady state error exists is ϕ_L . Hence, further tuning is needed and should be conducted with states estimated using Kalman filters.

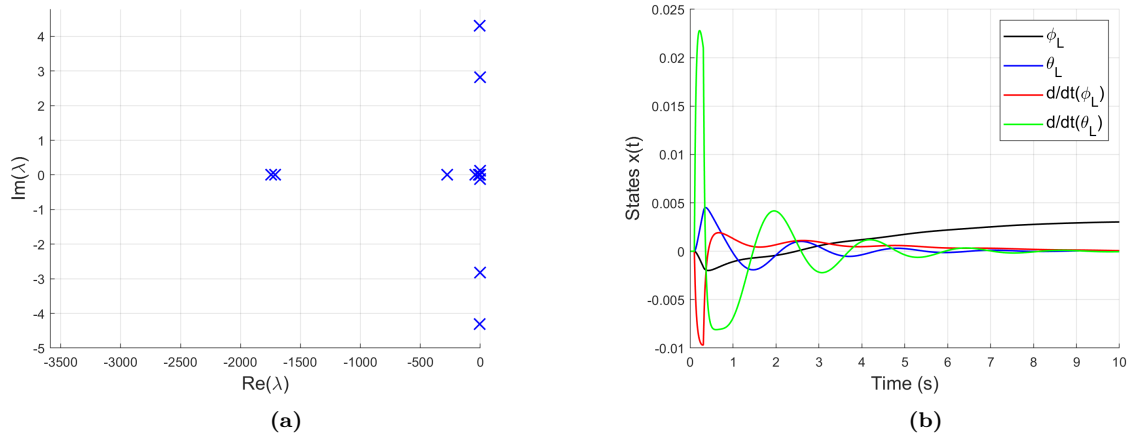


Figure 5: Shear force distribution over wing half span for symmetric loading case.

Table 2: Team Members and Roles.

State	Hover	Q_{ii}	Cruise	Q_{ii}
u (m/s)	± 1	1000	30	1000
v (m/s)	± 0.1	1000	± 0.1	1000
w (m/s)	± 0.1	10	± 0.1	10
p (rad/s)	± 0.1	1000	± 0.1	1000
q (rad/s)	± 0.1	10	± 0.1	10
r (rad/s)	± 0.1	1000	± 0.1	1000
Φ_e (rad)	± 0.01	100	± 0.01	1000
Θ_e (rad)	± 0.01	100	.04	100
Ψ_e (rad)	± 0.01	100	± 0.01	100
ϕ_L (rad)	± 0.1	10^4	± 0.01	100
θ_L (rad)	± 0.1	100	0.627	10^4
$\dot{\phi}_L$ (rad/s)	± 0.01	10^4	± 0.1	100
$\dot{\theta}_L$ (rad/s)	± 0.01	100	± 0.1	10^4

5.2 Planned Controller Evaluation tests

While LQR results in a highly stabilisable system, it has often been associated with poor robustness performance. Thus, any model mismatch or inaccuracy in predicting equilibrium points may adversely affect the system performance. To some extent, this is resolved with Kalman filtering as process noise and disturbances are modelled. To evaluate robustness, system stability will be tested when there is an offset in the initial state values, forcing the system to operate away from trim points. Sensitivity to state value offset is important to determine as realistically it may not be possible for the synchropter to operate at equilibrium points that it is designed for. Three tests are planned: Translatory velocity offset, euler angles offset and payload angle offset. Note that offset is applied to these three motions separately, so during velocity offset tests, all other states will kept at their trim values. This is to identify system sensitivity to perturbations in any particular degrees of freedom clearly. Settling time and overshoot for each state will be determined. Though heading angle Ψ_e has been included, it is not expected to alter system dynamics significantly. Selected offsets have been presented in the appendix.

6 Controller Design Challenges

6.1 Observability

Initially, the first 11 states of the state vector x (translational velocities, angular velocities and payload angles) were to be measured with sensors. Unfortunately, as demonstrated in (cite India), the system is not observable with this configuration. Further analysis shows that system could be observable if all payload states and yaw heading angle are measured (cite India). However, due to difficulties in measuring payload angular velocities with the current sensors and time constraints, this change could not be implemented, hence further tuning of the LQR controller and incorporation of the state estimator could not be completed.

7 Future Work

The presented design has plenty of scope for improvements and further development. Next step in the design process would be to successfully evaluate the Kalman filter gains and conduct an analysis of the closed loop response of the augmented system. Once controller tuning is complete, the evaluation tests recommended in 5.2 should be conducted to assess robustness. Additionally, currently the mission analysis is quite focused on helicopter motion in the longitudinal plane, but modelling sustained turns would present an interesting stability analysis. Finally, controller design should be extended to other mission legs, including autonomous legs with reference trajectory tracking to produce a robust, fully controllable vehicle.

For the second design iteration, several changes are recommended. Building up on suggested improvements in payload modelling mentioned earlier, control authority should be extended to directly control payload oscillations too. As mentioned previously, payload modes are undamped and their frequency of oscillation varies greatly with slung load properties. The method suggested in [12] elegantly uses delayed feedback and input shaping to predict payload oscillations and feedforwards predicted oscillations to the system. This results in an active control of payload oscillations through helicopter controls. Other simpler methods could involve just slightly altering the payload suspension point to a wider handle bar, which dampens lateral oscillations and reduces moments transferred

to the helicopter [13].

While the current LTI offline system is beneficial in terms of computational expense, it has several limitations. Firstly, large perturbations from trim points in any flight phase will not be modelled correctly, and hence the controller may not be able to stabilise the system. Additionally, some linearized system may not be controllable, but their nonlinear models are [10]. Moreover, several interesting mission legs cannot be modelled. For example, capture of the payload is a dynamic stage, with force being slowly added to the helicopter. This case will be critical for stability analysis, but this isn't possible with a time invariant model. If nonlinear models are generated, model predictive control is suggested. This is an online control mode which computes trajectory in real time while measuring system response. MPC is known to have excellent robustness properties, and has been used successfully for several autonomous applications.

Table 3: *Translatory velocity offset test values*

State	Hover	Cruise
u (m/s)	5	25
v (m/s)	5	5
w (m/s)	5	5

Table 4: *Euler angles offset test values*

State	Hover	Cruise
Φ_e (rad)	0.2	0.2
Θ_e (rad)	0.2	0.4
Ψ_e (rad)	0.2	0.2

Table 5: *Payload state offset test values*

State	Hover	Cruise
ϕ_L (rad)	0.1	0.1
θ_L (rad)	0.1	1
$\dot{\phi}_L$ (rad/s)	0.2	0.2
$\dot{\theta}_L$ (rad/s)	0.2	0.2

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A Appendices

A.1 Whatever you feel you need to add

Look at me i'm additional

B Supporting MATLAB Code

This script evaluates all of something and plots something

```
1
2 % AERO96005 GROUP DESIGN PROJECT
3 % HELICOPTER.TRANSMISSION.MLX
4 % -----
5 % DETERMINES REQUIRED GEAR RATIOS, EVALUATES TORQUE &
6 % TORQUE TUBE SIZING AND SEVERAL OTHER TRANSMISSION
7 % OBJECTS WHEN GIVEN THE REQUIRED ENGINE INPUT & ROTOR
8 % OUTPUT DETAILS.
9 % T CROSS 25/05/2020
10 % -----
11
12 % WORKSPACE BOOK-KEEPING
13
14 clear
15 clc
16
17 format longG
18
19 % conversions
```